

3.6 Intermolecular forces homework — answers

Section A — Multiple choice

A1 = C

A2 = C

A3 = B

A4 = B

A5 = B

A6 = A

A7 = B

A8 = D

Section B — Mark scheme clips

B1 [1]

1(b)(iii)	hydrogen bond	1
-----------	---------------	---

B2 [2]

1(b)(iv)	M1 link shown as a dashed line between the lone pair of electrons from N of one NH ₃ to one H on other NH ₃ M2 minimum 3 correct partial charges (on adjacent atoms) over two NH ₃ molecules EITHER $\delta^- \text{N} - \delta^+ \text{H} \cdots \delta^- \text{N}$ OR $\delta^+ \text{H} \cdots \delta^- \text{N} - \delta^+ \text{H}$	2
----------	---	---

B3 [2]

2(c)(iii)	N ₂ (g)	CO(g)	2
	instantaneous dipole–induced dipole ✓	instantaneous dipole–induced dipole (and) permanent dipole–permanent dipole ✓	

B4 [3]

1(d)(i)	SO ₂	–73	pd	3
	H ₂ O	0	H	
	SiO ₂	1600	C	
	Na ₂ O	1132	I	
	MgO	2852	I	

B5 [2]

1(a)(ii)	M1 bar at HF shows any boiling point above HI on graph M2 explanation of difference in boiling point of a sample of HF in terms of strength (sum of) hydrogen bonds (and induced dipoles in HF) are stronger than (sum of) induced dipoles (and permanent dipoles in HCl/ HBr/ HI)	2
----------	--	---